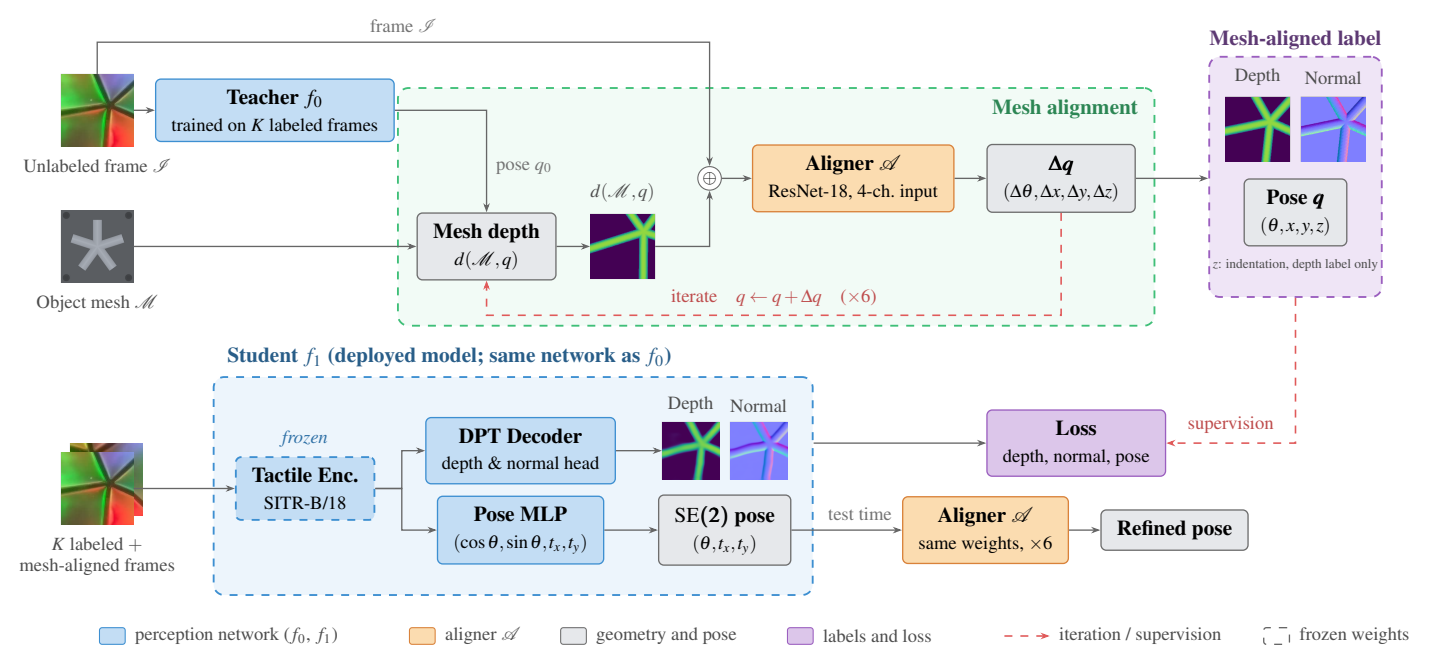

frame $\mathscr{I}$
Unlabeled frame $\mathscr{I}$
Teacher f_0
trained on K labeled frames
Object mesh $\mathscr{M}$
pose q_0
Mesh depth
$d(\mathscr{M},q)$
$d(\mathscr{M},q)$
Mesh alignment
Aligner $\mathscr{A}$
ResNet-18, 4-ch. input
Δq
$(\Delta\theta,\Delta x,\Delta y,\Delta z)$
iterate $q \leftarrow q+\Delta q$ $(\times 6)$
Mesh-aligned label
Depth
Normal
Pose q
(θ,x,y,z)
z: indentation, depth label only
Student f_1 (deployed model; same network as f_0)
K labeled + mesh-aligned frames
frozen
Tactile Enc.
SITR-B/18
DPT Decoder
depth & normal head
Pose MLP
$(\cos\theta,\sin\theta,t_x,t_y)$
Depth
Normal
SE(2) pose
(θ,t_x,t_y)
Loss
depth, normal, pose
supervision
test time
Aligner $\mathscr{A}$
same weights, ×6
Refined pose
perception network (f_0, f_1)
aligner $\mathscr{A}$
geometry and pose
labels and loss
iteration / supervision
frozen weights